

Electroweak Scale and Fermion Masses from Horizon Readouts

First-principles electroweak closure in `HQIV_LEAN`

Abstract

Derived first: from the null-lattice combinatorics and horizon-area functionals, the `HQIV_LEAN` formalization produces a closed electroweak sector without importing PDG inputs for it—a geometric vacuum $v = T_{\text{lockin}} S(\text{referenceM} + 1)(1 + \gamma)$ from the lock-in temperature and outer-horizon surfaces $S(m) = (m + 1)(m + 2)$, then gauge-boson and Higgs masses from v with triality-fixed couplings; the file proves rational witnesses (e.g. $M_W^{\text{derived}} = 392/5$ in the module’s units) and comparison theorems to electroweak reference values. The same backbone yields a derived neutrino ladder from $\gamma/S(\text{referenceM} + 2)$ applied to the neutral-vector mass. **Derived next:** a stars-and-bars lattice ratio forces $\alpha = \frac{3}{5}$ at every depth, and ratios of detuned $S(m)$ readouts drive generation steps and a rank- l^2 mass-scaling ansatz on content classes. The updated modal path now treats modal frequency / horizon data as primary through `ModalFrequencyHorizon.lean`, with lightweight charged-lepton ratios in `LeptonResonanceGlobalDetuning.lean` and outer-closure witnesses in `DerivedGaugeAndLeptonSector.lean`; natural indices remain evaluation coordinates. **Quarks and nucleons** use the same detuned-area ratios and lock-in binding spine, but still rely on explicit scale witnesses (heavy quark GeV anchors and integer readout triples); see Section 5. Fermion Planck-mass labels and further SM alignment are exported in `SM_GR_Unification` and companion modules; status is tracked in `AGENTS/MASS_DERIVATION_ROADMAP.md`.

1 The Core Derivation

(I) Electroweak scale from T_{lockin} and outer horizons. Let $m_{\text{lockin}} = \text{referenceM}$ and $T_{\text{lockin}} = T(m_{\text{lockin}}) = 1/(m_{\text{lockin}} + 1)$. `DerivedGaugeAndLeptonSector.lean` defines a geometric vacuum scale

$$v = T_{\text{lockin}} \cdot S(\text{referenceM} + 1) \cdot (1 + \gamma), \quad S(m) = (m + 1)(m + 2),$$

with $\gamma = 1 - \alpha = \frac{2}{5}$. Gauge and scalar masses are built from v with triality-normalized couplings (`su2CouplingDerived`, `u1CouplingDerived`). The file proves closed rational witnesses, e.g. `boson.witness_M.W`: $M_W^{\text{derived}} = 392/5$ (GeV-normalized units in the module), and companion theorems for M_Z and m_H , together with comparison theorems to PDG centrals (`raw_ew_boson.W.H.below_PDG_Z.above`, etc.). Neutrino masses begin with `m_nu_e_derived` equal to $(\gamma/S(\text{referenceM} + 2)) M_Z^{\text{derived}}$ with cascade to higher generations (`neutrino_masses_from_outer_horizon`). *No PDG literals are imported to define these objects.*

(II) Combinatorial imprint and horizon-surface readouts. `OctonionicLightCone.lean` proves a lattice ratio equals $\alpha = \frac{3}{5}$ for every truncation (`latticeAlphaRatio_eq_alpha`). `FanoResonance.lean` packages the shared surface $S(m)$ and detuned surfaces `detunedShellSurface`; resonance steps are ratios of these, hence a concrete “ $S(m)$ -octave” hierarchy feeding lepton and quark ladders (`ChargedLeptonResonance`, `QuarkMetaResonance`). Where a content-class ansatz is needed, `ConservedContentMa`

formulates scaling with a rank- l factor and shell correction (`massScalingAnsatz`), i.e. the expected l^2 weighting on top of the shell imprint—the combinatorial hierarchy is first-class, not an afterthought.

The current preferred lepton path is now:

`ModalFrequencyHorizon` $\rightarrow$ `LeptonResonanceGlobalDetuning` $\rightarrow$ `DerivedGaugeAndLeptonSector`,

with `ChargedLeptonResonance` retained as a thicker phenomenology/scratch layer. This reflects the updated project stance that horizons and effective surfaces are primary, while shell numbers are sampled readouts.

2 Maxwell spectrum + Δ triality spine (what the denominator stands on)

The denominator used in the active path is no longer a free rapidity stub; it is sourced from the matrix-level O-Maxwell/Fano scaffold in `Fano0maxwellSpectrum.lean`. The construction is:

1. **Carrier and line projector.** A Fano vertex is embedded into the octonion matrix carrier `Fin 8` as index 1..7 (scalar slot 0 reserved), then lifted to a diagonal selector. Summing selectors over a line L gives

$$\Pi_L = \sum_{v \in L} \Pi_v, \quad \text{tr}(\Pi_L) = 3$$

(theorem `fanoLineSelector_trace`).

2. **Phase-lift generator Δ .** `PhaseLiftDelta.lean` fixes the antisymmetric 8×8 generator with nonzero $(1, 7), (7, 1)$ entries; the scaffold proves

$$\text{tr}(\Delta \Delta^\top) = 2$$

(theorem `phaseLiftDeltaQuadraticTrace_eq_two`).

3. **Projection normalization.** The spectral scalar is defined by

$$\mathcal{N}(L) = \frac{\text{tr}(\Pi_L) + \text{tr}(\Delta \Delta^\top)}{5} = \frac{3 + 2}{5} = 1$$

(theorem `spectralProjectionNormalization_eq_one`).

4. **Projected mode strength and 1-jet.** For mode (L, m) , projected strength is $\mathcal{N}(L) \text{phaseLiftCoeff}(m)$, and the detuning source is

$$D_L(m) = 1 + \frac{3\gamma}{2} (\sigma_L(m) - \sigma_L(0)).$$

Using $\mathcal{N}(L) = 1$, `phaseLiftCoeff`(m) = $\varphi(m)/6$, and $\varphi(m) = 2(m + 1)$, Lean proves

$$D_L(m) = \text{rindlerDetuningShared}(m) = 1 + \frac{\gamma}{2} m$$

(`spectralFanoRindler1Jet_eq_rindler`, `spectralFanoRindler1Jet_eq_one_plus_half_gamma`).

The public denominator `trialityProjectedDenominator` in `FanoTrialityDetuningScaffold.lean` is exactly this spectral 1-jet, and `trialityProjectedDenominator.firstOrder` is therefore a proved corollary, not a placeholder.

What “triality” means here today. `Triality.lean` proves the $\text{Spin}(8)$ representation cycle (order-3 permutation of $8v, 8s^+, 8s^-$). What is *not* yet proved is the stronger forcing theorem that the denominator’s unit constant is uniquely imposed by triality-equivariant mode selection. So the current status is: *spectral Δ +Fano construction proved and affine-compatible; full triality-forcing of the constant term remains an open theorem objective.*

3 Backbone definitions (one pass)

`AuxiliaryField.lean`: $T(m) = 1/(m+1)$, $\varphi(m) = 2(m+1)$, `shell_shape`. `PhaseLiftDelta.lean`: fixed antisymmetric Δ generator and $\varphi/6$ phase-lift coefficient. `Triality.lean`: order-3 $\text{Spin}(8)$ representation cycle bookkeeping. `SM_GR_Unification.lean`: $\alpha_{\text{GUT}} = 1/42$ from cube directions $\times$ octonion imaginary units. `Baryogenesis.lean`: lock-in index `m_lockin` equals `referenceM`. `BoundStates.lean` / `HarmonicLadderMass.lean`: shell-resolved $\alpha_{\text{eff}}(m)$ and hydrogenic α^2 scalings.

4 Fermion and hadron exports (what `SM_GR_Unification` names)

`SM_GR_Unification.lean` exports Planck-unit masses via `smMassFromGeometry`: scale `m_tau_P1` divided by `resonanceProduct` (`ChargedLeptonResonance.lean`). Heavy triality slot has product 1, so τ , top, and bottom labels coincide at that functional’s value.

5 Quarks and nucleons: what is derived, what is still a witness

Honesty (not “no witnesses”). The quark ladder in `QuarkMetaResonance.lean` is *not* free of external numerals. In particular:

- **Absolute scale:** the up-type ladder is normalized by a named real `m_top_GeV` (PDG-style pole-mass listing used as calibration witness). Ratios between generations are *not* taken from PDG tables; they are fixed by `geometricResonanceStep` ratios built from detuned surfaces `detunedShellSurface` and explicit $\mathbb{N}$ shell endpoints (`m_quark_up_*`).
- **Readout tables:** the integer triples `m_quark_up_*` and `m_quark_down_*` are calibration inputs that determine internal resonance factors; uniqueness of these coordinates from the null lattice alone is *not* proved in this module.
- **Legacy down branch:** a separate real `m_bottom_GeV` still fixes the old down-type comparison ladder (`quarkMassDown`, `m_strange_GeV`, `m_down_GeV`) unless/until that path is replaced end-to-end by the active API.

Lock-in readout, `T_lockin`, and the GeV top anchor (disambiguation). The lock-in indices agree across modules: `m_lockin = referenceM = m_top_at_lockin` (`Baryogenesis`, `QuarkMetaResonance`), and

$$T_{\text{lockin}} = T(m_{\text{top_at_lockin}})$$

on the auxiliary ladder (`AuxiliaryField.lean`: $T(m) = T_{\text{P1}}/(m+1)$ with $T_{\text{P1}} = 1$ in natural units). `SM_GR_Unification` records this as `T_lockin_eq_temperature_at_quark_top_birth_shell`. So the **story** is exactly what you describe: lock-in, baryogenesis, and the “top birth” readout are the *same* calibration row, and `T_lockin` is the horizon temperature assigned to that row.

That is *not* the same proposition as “the real number `T_lockin` equals the pole mass m_t in GeV”: at `referenceM= 4` one has $T = 1/5$ in those units, not ~ 172 GeV. The explicit real `m_top_GeV` exists precisely to supply **laboratory GeV** normalization for the color-resonance ladder (`topLockinColorResonanceAnchorMass`, etc.); it is the witness that remains separate from the abstract ladder unless a future theorem closes the gap. A parallel export `m_top_quark_natural` from `smMassFromGeometryLabel.top` uses the τ Planck mass and resonance products only—no import of `m_top_GeV` there. ADM lapse and plasma corrections (`HQVM_lapse`; `QuarkMetaResonance` docstring) are *not* applied inside `T` itself, so “top energy” in the physical paper sense may require that layer; the Lean `T_lockin` identity is the lock-in temperature at that readout.

Why top is the heavy anchor (current implementation). The active color-composed API fixes the heavy up-like channel by `allowedColorResonanceMass.upLike.heavy = m_top_GeV`. This is not arbitrary bookkeeping: it matches the lock-in-heavy channel where the resonance product is 1, so the heavy band is the unique non-relaxed reference from which mid/light bands are generated by the same detuned-surface readout map. In other words, top is used as the anchor because the current ladder is implemented as “*heavy lock-in witness + horizon/surface relaxations*” rather than as three independently normalized flavor masses.

What *is* derived in-structure from the same spine as leptons ($S(m)$, Rindler detuning, γ , α , `referenceM`) are the **ratio factors** between shells and the **composite-trace** binding `nucleonSharedBinding_MeV` at `referenceM`.

Active visible-state API (color-composed ladder). The public masses `allowedColorResonanceMass` normalize the heavy up-like band to the top anchor, then compress the heavy down-like visible mass by a cross-channel detuning

$$\text{crossChannelHeavyShellDetuning} = \frac{\text{detunedShellSurface}(\text{m_quark_up_top_shell})}{\text{detunedShellSurface}(\text{m_quark_down_bottom_shell})}$$

and a visible bookkeeping factor `chargedLeptonContentCount×cubeAxes = 2 × 3` (two visible charge signs across three color channels). Mid/light bands descend by the same `downResonanceProduct / upResonanceProduct` machinery as elsewhere. Nucleon constituents use the active light masses `activeUpLightQuarkMass_GeV`, `activeDownLightQuarkMass_GeV` with a shared dressing scale and a smaller residual detuning tied to `alphaEffAtShell referenceM`; `DerivedNucleonMass.lean` exports `derivedProtonMass`, `derivedNeutronMass`.

Representative numerics (module units, current witness set). Evaluating the definitions with the literals in the repository (same formulas as `QuarkMetaResonance.lean`; approximate) gives:

charm (from top / first up-step)	≈ 1.27 GeV
active up-like light (<code>allowedColorResonanceMass upLike light</code>)	≈ 0.00220 GeV
active down-like heavy (compressed)	≈ 4.88 GeV
active down-like light	≈ 0.00550 GeV
<code>derivedProtonMass</code>	≈ 970.4 MeV
<code>derivedNeutronMass</code>	≈ 971.8 MeV
$m_n - m_p$	≈ 1.35 MeV

These are **not** PDG matches; they are the output of the closed Lean definitions given the current anchors and readout tables.

Why they differ from observation. PDG values for quarks and hadrons are quoted in different schemes (pole vs. $\overline{\text{MS}}$, running scales) and include nonperturbative QCD and finite-volume/lattice input where applicable. This formalization instead implements a **finite-chart resonance surrogate**: fixed $\mathbb{N}$ readout endpoints, a single heavy-quark GeV anchor, and a composite-trace binding ansatz at `referenceM`. Differences are therefore expected:

- **Proton** ~ 938 MeV **vs.** ~ 970 MeV **here**: the model does not claim a full nonperturbative nucleon mass calculation; the residual error reflects missing strong-sector closure and the simplicity of the dressing/binding split.
- **Light quark “masses”**: experimental light-quark masses are indirect and scheme-dependent; our GeV numbers are internal ladder coordinates, not a claim of pole masses for u, d .
- **Heavy flavors**: charm/bottom/top comparisons would require either removing the top anchor or deriving it from a deeper lock-in theorem; until then, agreement with PDG is not the target of the proof artifact.

6 Explicit inputs and proof obligations (AGENTS-aligned)

The file `AGENTS/MASS_DERIVATION_ROADMAP.md` is the authoritative status board. In short:

- **Faithful to the light-cone spine today**: bosonic closure witnesses and neutrino witnesses in `DerivedGaugeAndLeptonSector.lean`; structural binding quantities (`nucleonSharedBinding_MeV`, composite-trace packaging) tied to `referenceM`, α , γ , and $S(m)$.
- **Still carried by explicit numerals or boundary conditions**: substrate pins (`qcdShell`, `latticeStepCount` $\Rightarrow$ `referenceM`); **quark sector**: `m_top_GeV`, legacy `m_bottom_GeV`, and the $\mathbb{N}$ shell triples in `QuarkMetaResonance.lean` (Section 5); several SM alignment decimals in `SM_GR_Unification.lean` where the code matches paper fiducials by definition.
- **What the roadmap asks next**: reduce anchor count and unify the legacy down ladder with the active visible-state API—see the milestone table in `AGENTS/MASS_DERIVATION_ROADMAP.md`.

This division is intentional: the formalization *separates* closure witnesses derived from $(T_{\text{lockin}}, S, \gamma, \alpha)$ from numerals whose job is alignment or unfinished closure. Readers evaluating the program should use the roadmap’s honesty table, not footnotes in a survey note.

7 Octonions and the Standard Model bookkeeping layer

`SMEembedding.lean`, `Triality.lean`, `Generators.lean`: octonionic 8-dimensional carriers, $\mathfrak{so}(8)$ generators, and Standard Model quantum numbers. This layer fixes representation labels; the dynamical masses above come from the shell program in Sections 1–5.

Reading order in the repository

`OctonionicLightCone.lean` $\rightarrow$ `AuxiliaryField.lean` $\rightarrow$ `PhaseLiftDelta.lean` $\rightarrow$ `FanoLine.lean`
 $\rightarrow$ `OMaxwellAlgebraSeed.lean` $\rightarrow$ `FanoOMaxwellSpectrum.lean` $\rightarrow$ `FanoTrialityDetuningScaffold.lean`
 $\rightarrow$ `ModalFrequencyHorizon.lean` $\rightarrow$ `LeptonResonanceGlobalDetuning.lean` $\rightarrow$ `DerivedGaugeAndLeptonSector`
 $\rightarrow$ `FanoResonance.lean` / `QuarkMetaResonance.lean` (Section 5) $\rightarrow$ `DerivedNucleonMass.lean`

→ `SM_GR_Unification.lean`; optional thicker lepton phenomenology: `ChargedLeptonResonance.lean`;
parallel: `AGENTS/MASS_DERIVATION_ROADMAP.md`.

Acknowledgments

Mathlib supplies analysis and linear algebra. Maintainership notes: `AGENTS/ASSUMPTIONS.md`,
`AGENTS/THEOREMS.md`.